

EMILY YU

Assistant Professor, Leiden Institute of Advanced Computer Science, Leiden University

<https://emillyu.nl/> — Z.YU@LIACS.LEIDENUNIV.NL — 0000-0002-4993-773X

Address: Gorlaeus Building, Einsteinweg 55, 2333 CC Leiden, Netherlands

RESEARCH INTERESTS

Formal verification and automated reasoning, especially for hardware designs and AI-based systems;
Runtime monitoring and control: ensuring safety and correctness of complex dynamical systems.

EDUCATION

Johannes Kepler University Linz, Austria 2019 – 2023

Ph.D. in Computer Science *with distinction*

Thesis: *Certifying Hardware Model Checking*

Supervisor: Prof. Armin Biere

Imperial College London, United Kingdom 2014 – 2018

M.Eng in Computing, *First Class Honours*

Integrated Master's degree: 3-year Bachelor + 1-year Master

Thesis: *Model Checking Temporal Epistemic Logic under Bounded Recall*

Supervisor: Prof. Alessio Lomuscio

Concord College, United Kingdom 2012 – 2014

GCE A-levels (High School)

PROFESSIONAL EXPERIENCE

Leiden University, Netherlands Since Oct 2025

Tenured Assistant Professor

- Leiden Institute of Advanced Computer Science

Institute of Science and Technology Austria, Austria Sept 2023 – Sept 2025

Postdoctoral Researcher

- Working with Prof. Thomas A. Henzinger on runtime monitoring and neural control verification.

Woodpecker Technologies, Singapore Aug 2022 – Jan 2023

External Consultant

- Pre-silicon formal verification for chip designs.

Johannes Kepler University Linz, Austria Feb 2019 – Jun 2023

Project Assistant

- Formal Models and Verification Group, lead by Prof. Armin Biere.

Morgan Stanley, London, United Kingdom April – Sept 2017

Software Engineer

- Full-stack development during a 6-month industrial placement.

BlackRock, London, United Kingdom Jun – Aug 2016

Software Engineer Intern

- Worked on internal web platform development.

Imperial College London, United Kingdom

2015 – 2016

Teaching Assistant

- Part-time employment. Courses: Discrete Maths, Logic, and Reasoning about Programs.

Imperial College London, United Kingdom

Jul – Sept 2015

Research Assistant

- 20 hours per week. Group of Prof. Chris Hankin. Research on games and abstractions for cybersecurity.

CAREER BREAKS

Maternity leave 2021-2022, and 2026.

HONORS & AWARDS

- **Distinguished Paper Award**, Int. Conf. on Computer Aided Verification (CAV), 2025
- **FWF ESPRIT Fellowship**, Principal Investigator, 2025
RAVENS: Runtime Assurance and Verification of Neural Systems
Austrian Science Fund, EUR 346,505
Host Institute: Vienna University of Technology, Austria
- **Disruptive Idea Award**, Neuro-symbolic Systems (NeuS) Conference, 2025
DARPA award, USD 100,000

LANGUAGE PROFICIENCY

English (fluent), Cantonese (fluent), German (B2), Mandarin (native)

ACADEMIC SERVICES AND OUTREACH

Member of Program Committee

- AAAI Conference on Artificial Intelligence (AAAI), 2027;
- SETSS 2026 Spring School on Engineering Trustworthy Software Systems, 2026;
- Int. Conference on Computer Design (ICCD), 2026;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD), 2026;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD) Student Forum, 2026;
- Int. Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026);
- Int. Symposium of Electronics Design Automation (ISED), 2026;
- Post-AI Formal Methods Workshop, 2026;
- AAAI Conference on Artificial Intelligence (AAAI), 2026;
- European Conference on Artificial Intelligence (ECAI), 2025;
- Int. Joint Conference on Artificial Intelligence (IJCAI), 2025;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD), 2025;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD) Student Forum, 2025;
- Int. Conference on integrated Formal Methods (iFM), 2024.

Member of Artifact Evaluation Committee

- Int. Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI), 2024;
- Int. Symposium on Formal Methods (FM), 2024;
- Int. Symposium on Automated Technology for Verification and Analysis (ATVA), 2024.

Journal Referee

- Science of Computer Programming, 2025.
- Journal of Artificial Intelligence Research (JAIR), 2025
- Formal Aspects of Computing Journal, 2025.

Conference (sub-)Referee

- IEEE Int. Symposium on Circuits and Systems (ISCAS) 2026;
- Int. Conference on Neuro-symbolic Systems (NeuS), 2025;
- Int. Conference on Automated Deduction (CADE), 2025.

Scientific Referent for Funding Agencies

- French National Research Agency (ANR), 2025.

Event Organization

- CAV 2021, Virtual. A member of the volunteering team.
- QONFEST 2020, Vienna, Austria. A member of the volunteering team.
- Kurt Goedel Conference 2019, Vienna, Austria. A member of the volunteering team.

Other Committees

- Committee member of the Diversity Committee of the Leiden Institute of Advanced Computer Science (LIACS), Leiden University, since 2025.
- Committee member of the LIAXX (women's network) of the Leiden Institute of Advanced Computer Science (LIACS), Leiden University, since 2025.

PHD/POSTDOC SUPERVISION

Harshul Gupta (PhD student) at Leiden University, starting June 2026.

Yanni Dong (Postdoc) at Leiden University, May 2026 - Present.

STUDENT SUPERVISION

Bachelor's thesis supervisor at Leiden University for Enzo Beekman, Feb 2026-Present.

Thesis title: *Certifying Constraints for Hardware Model Checking*

Master's thesis supervisor at Leiden University for Guangyu Zhu, Feb 2026-Present.

Thesis title: *Signal First-Order Logic Neural Control*

Master's thesis supervisor at Leiden University for Zhengqi Lang, Oct 2025-Present.

Thesis title: *Robust Certificates for Neural Control*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Fabian Kresse, Spring 2025. Project topic: *Verification of Logic Gate Neural Networks*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Sergei Pankratov, Spring 2025. Project topic: *Certificate-based Neural Control*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Fabian Kresse, Winter 2024. Project topic: *Predictive Runtime Monitoring for Complex Dynamical Systems*

TEACHING EXPERIENCE

Lecturer, **Formal Methods for AI.**

Winter 2026

Mandatory course for CS Master's students in Foundations of Computing

[This is a self-designed course by myself.](#)

Leiden University, Netherlands

Lecturer, **Software Engineering.**

Spring 2026

Leiden University, Netherlands

Teaching Assistant, **Discrete Maths, Reasoning about Programs, Logic.**

2015 - 2016

Imperial College London, London, United Kingdom.

RESEARCH VISITS & OTHERS

June-July 2024, Prof. Joost-Pieter Katoen, RWTH Aachen University, Aachen, Germany.

June-July 2024, Prof. Jürgen Giesl, RWTH Aachen University, Aachen, Germany.

Since 2023, core contributor to an Intel funded project on certifying hardware model checking, proposed by Prof. Armin Biere.

Former member of the ERC project VAMOS (2023-2025); PI: Prof. Thomas A. Henzinger.

Core developer of certificate checkers used for hardware model checking competitions.

Former member of an EPSRC project titled Games and Abstractions (2015), under the supervision of Prof. Chris Hankin at Imperial College London.

ACADEMIC REFERENCES

Prof. Thomas A. Henzinger (Institute of Science and Technology Austria) – tah@ist.ac.at

Prof. Armin Biere (University of Freiburg, Germany) – biere@cs.uni-freiburg.de

Prof. Keijo Heljanko (University of Helsinki, Finland) – keijo.heljanko@helsinki.fi

Prof. Martina Seidl (Johannes Kepler University Linz, Austria) – martina.seidl@jku.at